Identification and Quantification of Flavor Attributes present in Chicken, Lamb, Pork, Beef, and Turkey

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Abstract: The objectives of this study were to use a meat flavor lexicon to identify and quantify flavor differences among different types of meats such as beef, chicken, lamb, pork, and turkey, and to identify and quantify specific flavor attributes associated with "beef flavor" notes. A trained descriptive panel with 11 participants used a previously developed meat lexicon composed of 18 terms to evaluate the flavor of beef, chicken, pork, turkey, and lamb samples. Results show that beef and lamb samples can be described by flavor attributes such as barny, bitter, gamey, grassy, livery, metallic, and roast beef. Inversely related to these samples were pork and turkey and those attributes that were closely related to them, namely brothy, fatty, salty, sweet, and umami. Chicken was not strongly related to the other types of meats or the attributes used. The descriptive panel also evaluated samples of ground beef mixed with chicken to identify and quantify flavor attributes associated with a "beef flavor." Meat patties for this portion consisted of ground beef mixed with ground chicken in varying amounts: 0%, 25%, 50%, 75%, and 100% beef, with the remainder made up of chicken. Beef and beef-rich patties (75% beef) were more closely related to flavor attributes such as astringent, bloody, fatty, gamey, metallic, livery, oxidized, grassy, and roast beef, while chicken was more closely associated with brothy, juicy, sour, sweet, and umami. This research provides information regarding the specific flavor attributes that differentiate chicken and beef products and provides the first set of descriptors that can be associated with "beefy" notes.

Keywords: beef, chicken, descriptive analysis, flavor lexicon, sensory evaluation

Potential Application: The use of a standardized flavor lexicon will allow meat producers to identify specific flavors present in their products. The impact is to identify and quantify negative and positive flavors in the product with the ultimate goal of optimizing processing or cooking conditions and improve the quality of meat products.

Introduction

One of the main applications of descriptive sensory techniques is to generate lexicons that provide qualitative and quantitative information about the attributes that characterize a product in terms of flavor or texture. Flavor lexicons are especially useful when slight changes in flavor attributes of a specific product result in different product quality and consumers' acceptability. These slight differences in flavor attributes can be originated by several factors such as changes in processing conditions and chemical composition.

Flavor lexicons have been used to identify and quantify flavor attributes in many food products such as cheese, chocolate, and wine (Murray and others 2001; Drake and Civille 2003). This technique has also been used in several meat products such as beef, lamb, and chicken. Crouse and others (1983) described the flavor profile of lamb using terms such as ammonia, astringent, bitter, browned, gamey, grassy, metallic, muttony, musty, sour, and sweet. In a study on warmed-over flavors, chicken was described using the terms cardboard-like, chicken meat, rancid, vegetable oil, bread, toasted, and nut-like, among others (Byrne and others 1999b). Similarly, the flavor of pork was described by several terms such as cooked pork, rancid, bread, vegetable oil, fish, and nut-like, among others (Byrne and others 1999a). The flavor characteristics of beef have been studied by many authors, many of which use the AMSA (1995) guidelines in their assessment. Although this guide is a good starting point, it is not all-inclusive in describing flavor attributes. Many studies have looked at flavors in beef, but not with the intention of developing a standardized lexicon that can be used in other meats (Carmack and others 1997; Rowe and others 2009). Johnson and Civille (1986) used a lexicon that included terms such as cooked beef lean, cooked beef fat, browned, serum/bloody, grainy/cowy, cardboardy, oxidized/rancid/painty, and fishy. Lynch and others (1986) described beef flavors using astringency/drying, beefiness, bloody/serumy/metallic/sharp, dairy/milky fat, freshness, stale/off, oily, metallic, sweet, sour, bitter, and salty. James and Calkins (2008) also evaluated beef quality using terms such as tenderness, amount of connective tissue, off-flavor, and juiciness. Stelzlini and Johnson (2008) evaluated beef for intensity of sensory off-flavor in general, in addition to identifying the presence of one of several descriptors such as metallic, grassy, livery, grainy, gamey, or other, though they did not rate the intensity of the descriptor. Stetzer and others (2008) evaluated tenderness, juiciness, saltiness,

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beef flavor, and oily mouthfeel. Other studies have also developed terms to describe beef flavors, though they all differ in terms from each other (Miller and others 1996; Luchsinger and others 1997; Yancey and others 2005; Lorenzen and others 2005). Sensory studies on other types of meats include tests performed in chicken (Zhuang and others 2007; Zhuang and Savage, 2008, 2011; Brannan, 2009), turkey (Bagorogoza and others 2001; Brunton and others 2002), lamb (Resconi and others 2009; Gasperi and others 2005; Young and others 2003), and pork (O'Sullivan and others 2002; Meinert and others 2007; Byrne and others 1999; Utrilla and others 2010) to name a few. The studies previously mentioned were all focused on developing and using terms that are specific to a single meat product, such as chicken or beef. The current study aims to compare multiple meat products using a previously developed lexicon (Maughan and others 2012). This lexicon was developed by our laboratory following procedural guidelines from the American Meat Science Association (1995) in an attempt to standardize a flavor lexicon for meats. The lexicon consists of 18 terms that describe positive and negative notes commonly found in meats.

The objectives of this study were to use our previously developed meat flavor lexicon (Maughan and others 2012) to identify and quantify flavor differences among different types of meats such as beef, chicken, lamb, pork, and turkey, and to identify and quantify specific flavor attributes associated with "beef flavor" notes.

Materials and Methods

Meat samples

All meat samples for this study were obtained from local grocery stores. Fresh boneless skinless chicken breasts (all white meat) were ground up using a commercial grinder (Hobart model 4152, The Hobart mfg co., Troy, Ohio, U.S.A.). Previously ground, unseasoned commodity beef chuck (80% lean), pork (70% lean), turkey breasts (all white meat), and lamb (80% lean) from local stores were also used in this study. Meat samples were used on the same day of purchase, and were at a minimum of 7 d before the "use by" date on the packaging. Samples were not branded, and were purchased from several different stores throughout the study to give a more representative sampling of that type of meat in general. The ground meat samples were formed into 100 g patties using a hand-held hamburger press. For the test on beef and chicken mixtures, the ground beef and chicken were mixed in different proportions by weight before being formed into patties. The samples included 100% ground beef (100:0), 75% beef and 25% chicken (75:25), 50% beef and 50% chicken (50:50), 25% beef and 75% chicken (25:75), and 100% chicken (0:100).

Dry heat cooking on electric griddles (Tilt'n Drain Big Griddle 07046, National Presto Inc., Eau Claire, Wis. U.S.A.) at 163 °C was used to cook the patties following guidelines from the American Meat Science Association (1995). Patties were monitored at their center for internal temperature by using an AquaTuff 35200 digital thermometer (Atkins Technical Inc, Gainesville, FL U.S.A.) with a fast-responding microneedle probe. The probe was inserted horizontally from the side of the patties to reach the center of the sample, and a minimum of two readings per sample were taken to verify that the samples had reached the target internal temperature of 70 °C. After cooking, samples were cut into 2.54 cm cubes, placed in covered aluminum dishes, and served to the panelists immediately. Aluminum dishes did not impart flavor changes in the meat as no metallic flavor was reported by the panelists during training and/or tasting. Panelists tasted the samples in random order with three-digit blinding codes under red colored lights to minimize bias.

Descriptive sensory evaluation

A sensory descriptive panel ($n = 11$) was recruited and selected from the local community to develop a flavor lexicon for meats as described by Maughan and others (2012). In short, panelists were recruited using local newspapers advertisement and flyers and were screened based on their ability to differentiate between basic tastes in both identification and intensity rankings (American Society for Testing and Materials, 1981). Panelists who passed basic screening were recruited to participate in the meat flavor descriptive panel. Panelists ranged in age from 18 to 60, with nine males and two females, though demographics are not expected to influence the ratings in a trained descriptive panel. After recruitment, chicken, beef, turkey, pork, lamb, and deer were used by the panel to identify the most important attributes present in meats. Panelists were then trained for a minimum of 50 h to develop the meat lexicon and to identify and quantify these attributes following procedures described previously in Maughan and others (2012). A 15-point category scale (Muñoz and Civille 1998) was used to quantify the intensity of the flavor attributes detected. Panelists and panel performance was evaluated using a graphical panelist evaluation software to ensure consistency and reproducibility (Nofima Marin & Technical University of Denmark, 2008). Panelists were considered to be trained based on their ability to rate samples consistently between replications, as well as their ability to rate samples close to the rest of the panel based on no significant differences in an ANOVA test. Terms in the lexicon include astringent, barny, bitter, bloody, brothy, browned, gamey, grassy, juicy, fatty, livery, metallic, oxidized (warmed-over flavor), roast beef, salty, sour, sweet, and umami.

Throughout the project, panelist performance was evaluated both individually and as a group for consistency and accuracy. Samples were presented to panelists in individual booths, with results collected by computer. Unsalted crackers and water were given to the panelists to cleanse their palates between samples. Panelists waited for 5 minutes between tasting each sample. After completing the samples presented in each day, the panelists waited 15 min before tasting a replicate of the samples in a different randomized order. Consistency between replicates was used to ensure the panelists did not experience sensory fatigue.

Following the training, panelists evaluated the test samples. Beef and chicken mixtures were tasted on 2 d, and the other meat samples were tasted on an additional 2 d. Panelists tasted half of the samples with a replicate on each day.

Statistical analysis

Statistical analysis with the proc glm function was used for analysis of variance (ANOVA) to identify statistically significant differences at the 95% confidence level in SAS (version 9.1.3, SAS Institute Inc., Cary, N.C., U.S.A.). Comparison of the unadjusted means was made by a linear regression analysis using proc glm with the REGWQ method to test for significance at $\alpha = 0.05$. Principal component analysis (PCA) using proc factor with proc corr was used to analyze the relationship of lexicon terms to the samples.

Results and Discussion

Application of the lexicon in different types of meats

The ability of the lexicon to evaluate different types of meat was tested through the evaluation of ground beef, chicken, lamb, pork, and turkey. The means of the panelist ratings for each meat type are shown in Table 1. Significant differences were found in every attribute besides bloody and oxidized.

Ground beef, chicken, and lamb all had significantly higher astringency notes than pork and turkey. Ground lamb had the highest amount of barny, an attribute that represents the skatole (or "manure") taste. This was an expected result since lamb is usually associated with barny flavors (Schreurs and others 2007). Beef was the only other meat that had barny present, but it was significantly lower than the lamb and not significantly different from the other samples. Browned attribute was significantly different between ground beef and chicken, while pork was rated similar to beef in brownness, and no differences were found between turkey and lamb. Gamey and grassy were both significantly higher in beef and lamb, while pork and turkey were both rated significantly higher in juicy. Pork received the highest scoring for fatty, followed by turkey, lamb, and beef, which were not significantly different among them. The lowest value of fatty was observed for the chicken samples. Similar to the results found for barny, lamb was rated significantly higher in livery. Metallic was present in low levels in most of the meats, and was significantly higher in beef and lamb. Brothy was highest in pork and turkey followed by beef and chicken, with the lowest value obtained for the lamb samples. The salty attribute was rated highest in pork and the intensity of this attribute decreased in the following order: turkey, chicken, beef, and lamb. Sweet and umami were both significantly higher in chicken, pork, and turkey compared to beef and lamb. Even though there were some significant differences between samples in the rating of bitter, livery, roast beef, and sour attributes, these were only present in small amounts in the meat samples.

Principal component analysis of the results shows a strong relationship between beef, lamb, and certain attributes such as barny, grassy, gamey, livery, and roast beef (Figure 1). Pork and turkey were closely related to each other and to attributes such as brothy, browned, juicy, fatty, and salty. Beef, lamb, and their related attributes were inversely related to pork, turkey, and their respective attributes. Chicken was not closely related to any other type of meat. As seen in Figure 1, 60.2% of the variability in the data was explained by the horizontal axis (principal component 1), while 22.5% was explained by the vertical axis (principal component 2).

Previous studies on the flavor characterization of meats show that chicken flavors can be mainly described by descriptors such as salty, fatty, brothy, and cardboardy (oxidized) (Brannan, 2009; Zhuang and Savage 2011, 2008; Zhuang and others 2007), while lamb flavors can be described with attributes such as livery, oxidized, barny, and metallic (Resconi and others 2009; Gasperi and others 2005; Young and others 2003), and pork flavors can be described with attributes such as umami, sour, cardboard (oxidized), salty, fatty, and roast taste (Meinert and others 2007; Byrne and others 1999a; Utrilla and others 2010). These results are in accordance with the data reported by our research. It is important to note here that several processing conditions such as diet, storage time, cooking method, etc. can affect the taste within a specific type of meat. For example, in our previous research we showed that beef flavor can be affected by cattle's diet (Maughan and others 2012).

Data obtained from our research suggest that the flavor of chicken, pork, beef, turkey, and lamb can be differentiated with the flavor lexicon developed by our laboratory. This is especially significant for the chicken compared to the other meat samples.

Descriptive analysis of beef and chicken mixtures

The descriptive panelists used the lexicon to evaluate beef and chicken mixtures in an attempt to identify the flavors typically associated with the red meat "beefy" flavor. As described in the Materials and Methods section, five mixtures of beef and chicken were evaluated, including 100% beef (100:0), 75% beef and 25% chicken (75:25), 50% beef and 50% chicken (50:50), 25% beef and 75% chicken (25:75), and 100% chicken (0:100). One of the primary goals of this study was to identify specific flavors that characterize beef products. Chicken was added to beef as described before since the flavor characteristics of chicken were not correlated to the ones found in beef as shown in Figure 1. The average panelist ratings can be seen in Table 2. The panel found significant differences ($p < 0.05$) between the samples in astringent, brothy, fatty, gamey, grassy, juicy, metallic, oxidized, salty, sweet, and umami. Table 2 also shows that there are trends as the amount of chicken increased in the sample: astringent and fatty decreased in intensity as the amount of chicken increased (from 1.86 to 0.23 and 3.11 to 1.91, respectively). Juicy, brothy, salty, sweet, and umami increased (from

Table 1–Descriptive flavor ratings of various types of meat, including beef, chicken, pork, turkey, and lamb. Values reported are mean values ± standard errors. Samples with the same superscript letter are not significantly different at $P < 0.05$. A 15-point category scale was used to rate flavors intensity.

Attribute	Ground beef	Ground chicken	Ground pork	Ground turkey	Ground lamb	*P*-value
Astringent	1.67 ± 0.24^{a}	2.43 ± 0.30^{a}	0.12 ± 0.04^{b}	0.35 ± 0.07^{b}	1.67 ± 0.18^{a}	0.0001
Barny	1.08 ± 0.23^{b}	0.00 ± 0.00^{b}	0.08 ± 0.04^{b}	0.04 ± 0.02^{b}	2.88 ± 0.41^{a}	0.0001
Bitter	0.29 ± 0.07^{ab}	0.21 ± 0.07^{b}	0.06 ± 0.03^{b}	0.10 ± 0.02^{b}	0.54 ± 0.09^{a}	0.0075
Bloody	0.27 ± 0.06	0.12 ± 0.04	0.08 ± 0.04	0.35 ± 0.08	0.31 ± 0.08	0.2832
Browned	1.04 ± 0.13^{a}	0.21 ± 0.06^{b}	1.29 ± 0.15^{a}	0.69 ± 0.11^{ab}	0.69 ± 0.12^{ab}	0.0054
Gamey	1.46 ± 0.27^{a}	0.04 ± 0.02^{b}	0.00 ± 0.00^{b}	0.12 ± 0.04^{b}	1.62 ± 0.23^{a}	0.0001
Grassy	0.46 ± 0.09^{a}	0.04 ± 0.02^{b}	0.00 ± 0.00^{b}	0.12 ± 0.04^{b}	0.62 ± 0.11^{a}	0.0001
Juicy	0.87 ± 0.13^{b}	0.67 ± 0.11^{b}	3.15 ± 0.30^{a}	2.58 ± 0.30^{a}	1.12 ± 0.15^{b}	0.0001
Fatty	2.85 ± 0.31^{b}	1.29 ± 0.14^{c}	6.57 ± 0.32^{a}	3.44 ± 0.29^{b}	2.46 ± 0.22^{b}	0.0001
Livery	0.06 ± 0.03^{b}	0.31 ± 0.09^{b}	0.00 ± 0.00^{b}	0.12 ± 0.06^{b}	2.31 ± 0.32^{a}	0.0001
Metallic	1.38 ± 0.23^{a}	0.54 ± 0.13^{bc}	0.04 ± 0.27^{c}	0.23 ± 0.06^{c}	1.29 ± 0.16^{ab}	0.0016
Brothy	1.02 ± 0.16^{c}	1.69 ± 0.20^{bc}	2.38 ± 0.27^{ab}	2.69 ± 0.25^{a}	0.90 ± 0.14^{c}	0.0001
Oxidized	0.79 ± 0.13	0.79 ± 0.18	0.54 ± 0.16	0.25 ± 0.10	0.58 ± 0.11	0.5331
Roast Beef	0.52 ± 0.09^{a}	0.00 ± 0.00^{c}	0.12 ± 0.03^{bc}	0.27 ± 0.07^{abc}	0.38 ± 0.08^{ab}	0.0139
Salty	1.37 ± 0.15^{cd}	1.88 ± 0.14^{bc}	5.70 ± 0.31^{a}	2.40 ± 0.24^{b}	0.77 ± 0.11^{d}	0.0001
Sour	0.54 ± 0.12^{ab}	0.96 ± 0.13^{a}	0.12 ± 0.04^{b}	0.13 ± 0.04^{b}	0.54 ± 0.11^{ab}	0.0020
Sweet	0.23 ± 0.05^{b}	0.94 ± 0.14^{a}	1.42 ± 0.21^{a}	1.42 ± 0.17^{a}	0.10 ± 0.03^{b}	0.0001
Umami	2.52 ± 0.17^{b}	4.21 ± 0.22^{a}	4.54 ± 0.21^{a}	4.75 ± 0.25^{a}	2.73 ± 0.20^{b}	0.0001

1.43 to 3.80; 1.66 to 3.07; 1.32 to 2.89; 0.36 to 1.89, and 3.02 to 5.43, respectively) in intensity as the amount of chicken increased. These trends were not necessarily significant between each sample, but they were significant between the 100% beef and 100% chicken samples.

The descriptive panel results were also analyzed by principal component analysis, as seen in Figure 2. Principal component 1 in this plot contributes to 68.7% of the variability of the data, while principal component 2 contributes to 22.2% of the variability of the data, for a total of 90.9% of the variability. The PCA graph confirms what the statistical analysis showed in Table 2 that the amount of beef in the patties is strongly correlated with attributes such as bloody, livery, gamey, oxidized, grassy, astringent, fatty, and roast beef. As the amount of beef decreases in the mixture the correlation with these attributes decreases and samples become more related to attributes such as brothy, juicy, sour, sweet, salty, and umami. PCA analysis also shows a significant correlation among the different attributes, which are shown in Table 3. Strong trends in the flavor characteristics of beef/chicken samples when their composition shifts from chicken to beef are evident from the PCA

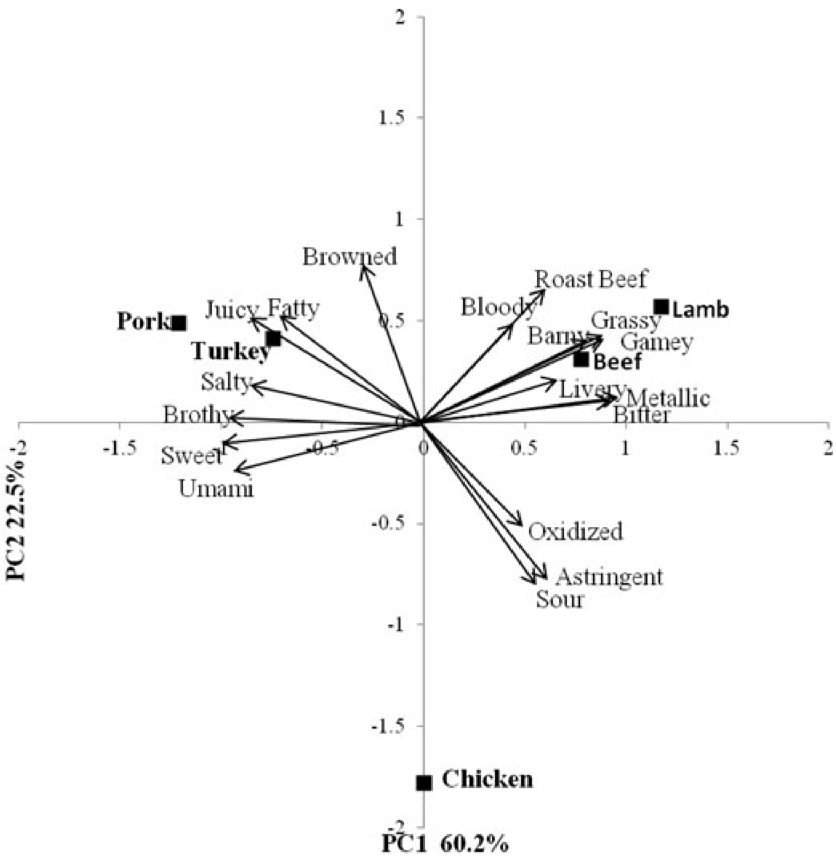

Figure 1–Principal component analysis of different types of meat, using the flavor lexicon.

Table 2–Mean descriptive panelist ratings for the beef and chicken mixtures. 0%, 25%, 50%, 75%, and 100% beef in chicken patties. Values reported are mean values ± standard errors. Samples with the same superscript letter are not significantly different at $P <$ 0.05. A 15-point category scale was used to rate flavors intensity.

Attribute	0%	25%	50%	75%	100%	*P*-value
Astringent	0.23 ± 0.05^{b}	0.43 ± 0.07^{b}	0.50 ± 0.10^{b}	0.75 ± 0.11^{b}	1.86 ± 0.20^{a}	0.0001
Barny	0.18 ± 0.06	0.00 ± 0.00	0.00 ± 0.00	0.05 ± 0.01	0.20 ± 0.11	0.1121
Bitter	0.32 ± 0.08	0.14 ± 0.04	0.14 ± 0.03	0.14 ± 0.05	0.32 ± 0.13	0.4474
Bloody	0.18 ± 0.06	0.18 ± 0.09	0.05 ± 0.02	0.30 ± 0.09	0.77 ± 0.33	0.0526
Brothy	3.07 ± 0.27^{a}	2.41 ± 0.17^{ab}	2.23 ± 0.16^{ab}	2.20 ± 0.17^{ab}	1.66 ± 0.11^{b}	0.0470
Browned	0.32 ± 0.08	0.75 ± 0.14	0.86 ± 0.15	0.55 ± 0.09	0.41 ± 0.14	0.1955
Fatty	1.91 ± 0.23^{b}	2.50 ± 0.31^{ab}	2.91 ± 0.29^{a}	2.95 ± 0.28^{a}	3.11 ± 0.30^{a}	0.0287
Gamey	0.00 ± 0.00^{b}	0.00 ± 0.00^{b}	0.00 ± 0.00^{b}	0.02 ± 0.01^{b}	0.75 ± 0.14^{a}	0.0001
Grassy	0.05 ± 0.02^{b}	0.00 ± 0.00^{b}	0.25 ± 0.08^{b}	0.57 ± 0.17^{b}	1.52 ± 0.26^{a}	0.0010
Juicy	3.84 ± 0.36^{a}	3.80 ± 0.30^{a}	2.36 ± 0.22^{b}	2.45 ± 0.20^{b}	1.43 ± 0.15^{b}	0.0001
Livery	0.00 ± 0.00	0.05 ± 0.02	0.00 ± 0.00	0.02 ± 0.01	0.09 ± 0.05	0.2483
Metallic	0.18 ± 0.06^{b}	0.14 ± 0.07^{b}	0.27 ± 0.07^{b}	0.30 ± 0.07^{b}	0.66 ± 0.12^{a}	0.0363
Oxidized	0.36 ± 0.08	0.66 ± 0.18^{b}	0.23 ± 0.07^{b}	0.61 ± 0.13^{b}	1.36 ± 0.21^{a}	0.0156
Roast Beef	0.36 ± 0.10	0.45 ± 0.12	0.86 ± 0.14	0.82 ± 0.14	0.84 ± 0.13	0.2790
Salty	2.89 ± 0.16^{a}	2.52 ± 0.17^{ab}	2.41 ± 0.18^{ab}	1.93 ± 0.13^{bc}	1.32 ± 0.16^{c}	0.0003
Sour	0.70 ± 0.13	0.57 ± 0.09	0.57 ± 0.11	0.32 ± 0.08	0.43 ± 0.08	0.4296
Sweet	1.89 ± 0.25^{a}	1.82 ± 0.24^{a}	1.36 ± 23^{ab}	0.82 ± 0.16^{bc}	0.36 ± 0.10^{c}	0.0001
Umami	5.43 ± 0.26^{a}	5.23 ± 0.21^{ab}	4.61 ± 0.24^{bc}	4.23 ± 0.16^{c}	3.02 ± 0.29^{d}	0.0001

plot (Figure 2). As the samples proceed from chicken to beef, they are positioned from the left to the right on the horizontal axis of the PCA plot. The 75:25 mixture was the sample that was most closely correlated with the beef sample on the right side of the plot, while the other beef samples were on the left side and therefore associated with chicken flavors. There is a clear delineation between the samples in the principal component analysis plot, showing that the panelists were able to clearly distinguish between the samples and identify the change in flavors that occurred between samples.

These tendencies are also evident from the ratings reported in Table 2. Attributes that were rated significantly different among samples were plotted as a function of percent of beef added and analyzed using linear regression (Figure 3). As seen in Figure 3, many of the attributes had a good linear fit with increasing beef percentage. A good linear relationship was found for astringent ($R^2 = 0.7687$), brothy ($R^2 = 0.8926$), fatty ($R^2 = 0.8679$), grassy ($R^2 = 0.7908$), juicy ($R^2 = 0.8912$), metallic ($R^2 = 0.7379$), salty ($R^2 = 0.9472$), sweet ($R^2 = 0.9563$), roast beef ($R^2 = 0.7627$), and umami ($R^2 = 0.9222$), while a poor relationship was found for gamey ($R^2 = 0.5200$) and oxidized ($R^2 = 0.4963$). Gamey was only present in the 100% beef sample, with more than 75% of beef needed in the mixture to detect this flavor; while oxidized did not follow a consistent pattern between all samples, making it difficult to fit a linear regression equation to either attribute. The strong fit of the linear regression equations confirms that there are

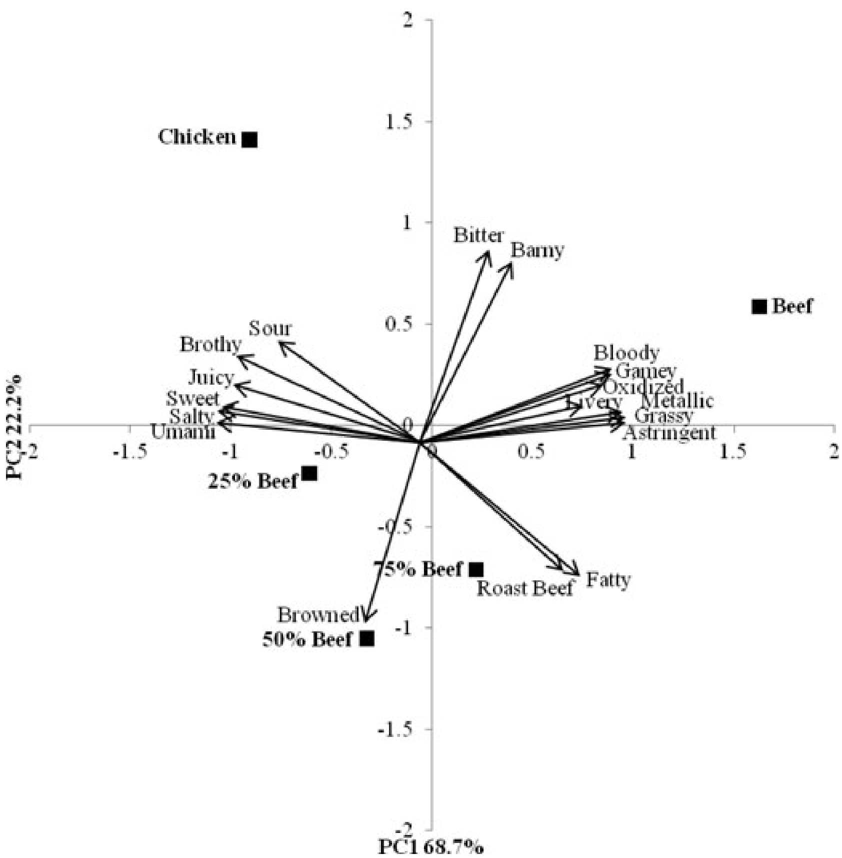

Figure 2–Principal component analysis of the beef and chicken mixtures.

Table 3–Correlation values for descriptive attributes used in the test.

	Astringent	Barny	Bitter	Bloody	Brothy	Browned	Fatty	Gamey	Grassy	Juicy	Livery	Metallic	Oxidized	Roast	Salty	Sour	Sweet	Umami
Astring	1.00	0.50	0.41	0.95*	−0.86	−0.30	0.70	0.96*	0.99*	−0.85	0.84	0.98*	0.9*	0.60	−0.95*	−0.59	−0.90*	−0.96*
Barny	0.50	1.00	0.98*	0.68	0.00	−0.92*	−0.20	0.65	0.56	−0.23	0.35	0.57	0.53	−0.13	−0.30	0.09	−0.33	−0.39
Bitter	0.41	0.98*	1.00	0.58	0.09	−0.86	−0.31	0.61	0.45	−0.12	0.31	0.49	0.45	−0.25	−0.16	0.29	−0.17	−0.27
Bloody	0.95*	0.68	0.58	1.00	−0.69	−0.55	0.48	0.95*	0.94*	−0.67	0.86	0.91*	0.97*	0.36	−0.87	−0.50	−0.81	−0.86
Brothy	−0.86	0.00	0.09	−0.69	1.00	−0.19	−0.95*	−0.73	−0.82	0.89*	−0.72	−0.81	−0.72	−0.82	0.93*	0.74	0.87	0.91*
Browned	−0.30	−0.92*	−0.86	−0.55	−0.19	1.00	0.35	−0.42	−0.38	0.02	−0.20	−0.35	−0.40	0.29	0.17	−0.02	0.21	0.20
Fatty	0.70	−0.20	−0.31	0.48	−0.95*	0.35	1.00	0.51	0.69	−0.88*	0.48	0.68	0.50	0.93*	−0.85	−0.81	−0.84	−0.83
Gamey	0.96*	0.65	0.61	0.95*	−0.73	−0.42	0.51	1.00	0.94*	−0.74	0.85	0.95*	0.92*	0.42	−0.84	−0.36	−0.77	−0.88
Grassy	0.99*	0.56	0.45	0.94*	−0.82	−0.38	0.69	0.94*	1.00	−0.88*	0.75	0.99*	0.87	0.64	−0.95*	−0.62	−0.94*	−0.98*
Juicy	−0.85	−0.23	−0.12	−0.67	0.89*	0.02	−0.88*	−0.74	−0.88*	1.00	−0.48	−0.90*	−0.59	−0.92*	0.90*	0.68	0.94*	0.95*
Livery	0.84	0.35	0.31	0.86	−0.72	−0.20	0.48	0.85	0.75	−0.48	1.00	0.72	0.96*	0.22	−0.76	−0.41	−0.60	−0.70
Metallic	0.98*	0.57	0.49	0.91*	−0.81	−0.35	0.68	0.95*	0.99*	−0.90*	0.72	1.00	0.84	0.65	−0.92*	−0.53	−0.91*	−0.97*
Oxidized	0.92*	0.53	0.45	0.97*	−0.72	−0.40	0.50	0.92*	0.87	−0.59	0.96*	0.84	1.00	0.29	−0.85	−0.50	−0.73	−0.80
Roast	0.60	−0.13	−0.25	0.36	−0.82	0.29	0.93*	0.42	0.64	−0.92*	0.22	0.65	0.29	1.00	−0.74	−0.73	−0.82	−0.78
Salty	−0.95*	−0.30	−0.16	−0.87	0.93*	0.17	−0.85	−0.84	−0.95*	0.90*	−0.76	−0.92*	−0.85	−0.74	1.00	0.80	0.97*	0.98*
Sour	−0.59	0.09	0.29	−0.50	0.74	−0.02	−0.81	−0.36	−0.62	0.68	−0.41	−0.53	−0.50	−0.73	0.80	1.00	0.82	0.71
Sweet	−0.90*	−0.33	−0.17	−0.81	0.87	0.21	−0.84	−0.77	−0.94*	0.94*	−0.60	−0.91*	−0.73	−0.82	0.97*	0.82	1.00	0.98*
Umami	−0.96*	−0.39	−0.27	−0.86	0.91*	0.20	−0.83	−0.88	−0.98*	0.95*	−0.70	−0.97*	−0.80	−0.78	0.98*	0.71	0.98*	1.00

*indicates significance at $P < 0.05$.

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trends for the changes in attributes that occur as the amount of beef increases, as previously seen in the PCA plot.

As shown in both the PCA and the linear regression plots, beef was closely related to such attributes as roast beef, fatty, astringent, grassy, gamey, and bloody. These are the core flavors that define beef using the developed lexicon. Although some of these could be considered potentially negative attributes, the plot shows that the attributes are only most strongly correlated with beef, as opposed to the other attributes which are more closely correlated with chicken. This does not mean that beef does not have other flavors, but rather that chicken had higher levels of those flavors and was more closely associated with attributes such as umami, brothy, juicy, sweet, and sour. These results are in accordance to previous data (Zhuang and others 2007; Zhuang and Savage 2008, 2011; Brannan 2009) reporting attributes such as brothy, sweet, salty, sour, and meaty associated with chicken flavors.

The data also show that there is a strong relationship between type of meat and their flavor characteristics. In addition to determining what flavor changes might be expected between all beef and all chicken patties, changes that occur with mixtures of the two meats can be seen as well. This could have an impact for meat producers who want to change the flavor characteristics of their products. For example, beef and chicken mixtures are common in hot dogs. If a producer found through descriptive techniques that their hot dogs had a gamey or grassy flavor, they might consider adding a higher proportion of chicken to decrease the intensity of those flavors. Additionally, they might also add more chicken to get a higher umami flavor, or more juiciness in their hot dog. The meat could then be profiled again by a descriptive panel and

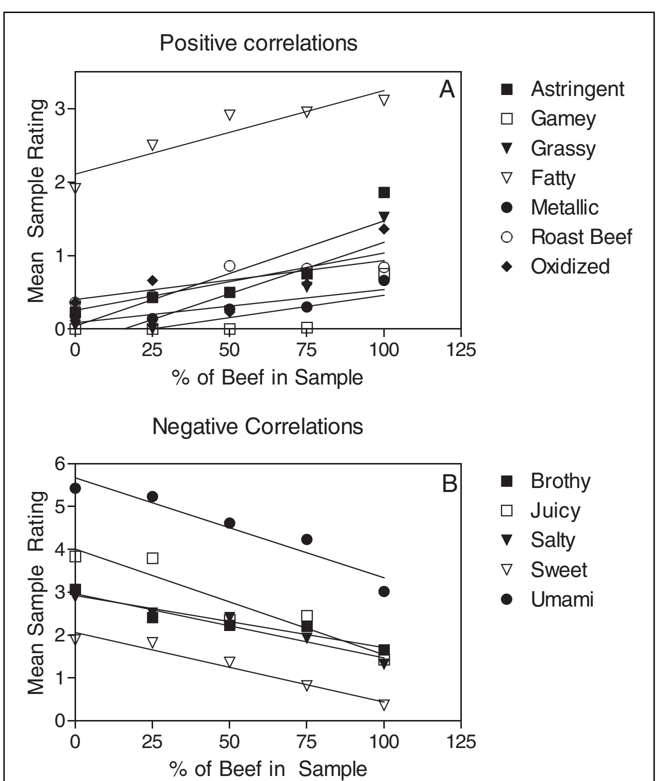

Figure 3–Linear correlations of significant attributes with increasing beef in the sample, based on the average panelist rating. (A) Positive correlations and (B) negative correlations.

paired with a consumer panel to determine if these changes in flavor characteristics were desirable from a consumer perspective.

Although there are a myriad of studies that examine the flavors of individual meat products, such as chicken or beef, there are not any studies that relate different types of meat to each other. Thyis study is an exploratory examination into this subject, with the goal of being able to evaluate and compare multiple types of meat using only one group of terms and scale. There are certain limitations inherent in this study, such as only one type of muscle or product representing each meat type. Future research, however, will allow for the inclusion of more types of meat, and will result in the refining of the terms used to increase their usefulness in comparing meat products.

Conclusions

A descriptive panel was used to evaluate the flavor differences between various types of meat, including beef, chicken, lamb, pork, and turkey. Beef and chicken mixtures were also evaluated by the panelists. Clear distinctions in flavor characteristics of the different types of meat were found by the descriptive panel. Beef and lamb were most closely related to flavor attributes such as roast beef, grassy, gamey, barny, livery, metallic, and bitter. Pork and turkey were inversely related to these, and were more closely related to juicy, fatty, salty, brothy, sweet, and umami notes. Chicken was not strongly related to any meats or attributes when all the meats were considered together.

Using the developed lexicon, beef was found to be more closely associated with astringent, bloody, oxidized, livery, fatty, gamey, grassy, and roast beef. Chicken was more closely associated with brothy, juicy, sour, sweet, and umami. These associations can be used in the industry combined with the developed lexicon to improve products that utilize both chicken and beef. Further research will increase understanding of the role that these changes in flavor play in consumer acceptance of these products. This is the first set of data that provides specific flavor attributes associated with "beefy" notes that are usually used to describe beef products.

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